

Theoretical and Experimental Investigation of Graphene/High- κ /p-Si Junctions

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Abstract—Here, we theoretically and experimentally investigate the impact of a high- κ layer inserted between graphene and p-Si in a graphene/Si junction. We have achieved 86-fold and 222-fold reductions in a specific contact resistivity (ρ_c) by inserting 1-nm-thick Al_2O_3 and 2-nm-thick TiO_2 in the graphene–semiconductor junction, respectively, corresponding to lowering the effective barrier height by 0.24 and 0.12 eV. Furthermore, we propose a graphene-induced gap state model that simultaneously considers the graphene’s modulation by a gate bias and the effect of the high- κ insertion.

Index Terms—Graphene, high- κ , Schottky, GS junction.

I. INTRODUCTION

SINCE THE beginning of a graphene era, the two-dimensional material has been used in various device applications, such as photodetectors [1], sensors [2], high-speed transistors [3], and energy devices [4], due to its superior electrical, mechanical, and optical properties. In particular, an important achievement was the development of the graphene barristor device, in which the graphene’s Fermi level (E_{FG}) is modulated in the source electrode instead of a channel region, requiring only the very low operating voltage of ~ 0.1 V [5]. However, a pinning phenomenon of the E_{FG} into a certain energy level is thought to degrade the performance of the barristor by increasing contact resistance (R_C) in the graphene–semiconductor (GS) junction and also by decreasing the degree of E_{FG} modulation by gate biasing (V_{GS}). Although interface passivation in the GS junction by means of hydroperoxide and sulfide treatment has been reported, no resulting enhancement in R_C or E_{FG} modulation was observed. Herein we theoretically and experimentally investigate the impact of inserting a high- κ layer between the graphene and Si in the GS junction, to confirm the possibility of improving the R_C and the effective

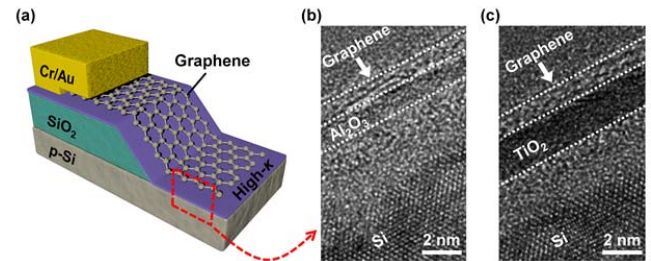


Fig. 1. (a) Schematic illustration of the GIS junction diode with a high- κ layer between the graphene and the p-Si. Cross-sectional TEM images of GIS junctions with inserted (b) Al_2O_3 and (c) TiO_2 layers.

modulation of E_{FG} by means of this technique. Moreover, we propose a graphene-induced gap state (GIGS) model that simultaneously considers the modulation of E_{FG} by V_{GS} and the effect of the high- κ insertion (i.e., the E_{FG} shift).

II. FABRICATION

Graphene was grown on Cu foil by a conventional chemical vapor deposition method. To fabricate GS junction diodes, we prepared SiO_2 (90 nm)/Si ($\sim 10^{17} \text{ cm}^{-3}$) substrates and then selectively etched the SiO_2 layer by means of a buffered oxide etch. Just after the etching step, a layer of high- κ material was deposited onto each patterned SiO_2 /Si substrate by means of plasma-enhanced atomic layer deposition (PEALD); the two used high- κ materials were Al_2O_3 and TiO_2 , and the deposited layers of these materials were 1 and 2 nm thick. This was confirmed through cross-sectional transmission electron microscopy (TEM) analysis. The grown graphene was transferred onto the samples by using a wet transfer method, and it was patterned by lithography and oxygen plasma etching processes. The patterned graphene samples were annealed at 300 °C for 1 hour in hydrogen/argon ambient to remove the polymer residue from graphene. During the annealing process, interfacial traps between Si and the high- κ layer would be also passivated. Finally, Cr/Au (10/50 nm) layers were deposited onto the graphene by an e-beam evaporator. The resulting graphene–insulator–semiconductor (GIS) junction diode structure is illustrated schematically in Fig. 1(a); Fig. 1(b) and (c) show the cross-sectional TEM images of fabricated GIS junctions, where continuous 1-nm Al_2O_3 and 2-nm TiO_2 layers can be found between Si and Cr/Au, respectively. The regions under high- κ layers appear to have been damaged during PEALD deposition. In addition, to fabricate conventional metal–insulator–semiconductor (MIS) junctions, three kinds of metals (Ti, Cr, and Au) were deposited on p-Si.

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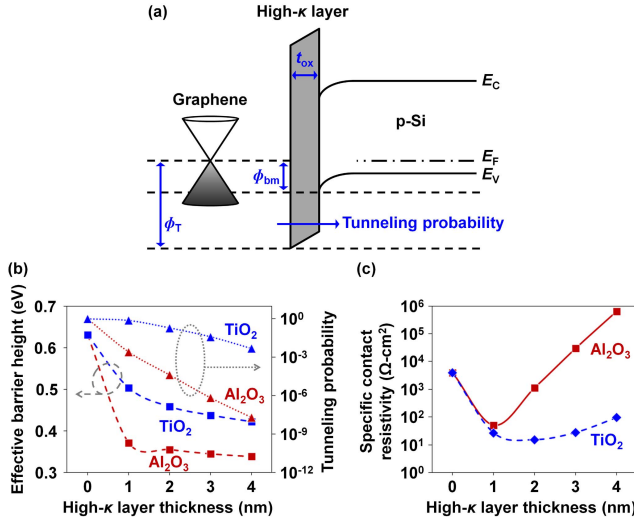


Fig. 2. (a) Energy band diagram of GIS junction with a high- κ layer. (b) Effective barrier height and tunneling probability versus high- κ layer thickness; (c) specific contact resistivity versus high- κ layer thickness.

Finally, current density–voltage (J – V) characteristics of the GS and MS junction diodes both with and without the high- κ layer were measured. In particular, temperature–dependent electrical measurements of GIS junction diodes were used to extract the effective hole barrier height between graphene and p-Si.

III. RESULT/DISCUSSIONS

Fig. 2(a) shows the energy band diagram of the GIS junction. Here, a carrier transport is expected to mainly obey a thermionic emission mechanism like that occurring at a conventional MS junction. However, because a high- κ layer is present between the graphene and Si, the effective barrier height (Φ_{bm}) and the probability of tunneling through this high- κ layer need to be considered in the GIS junction current equation; this equation can be written as

$$I = AA^*T^2\alpha \exp\left(\frac{-q\Phi_{bm}}{k_B T}\right) \left[\exp\left(\frac{qV_a}{\eta_{id}k_B T}\right) - 1\right], \quad (1)$$

where A is the junction area, A^* is the effective Richardson constant, α is the tunneling probability, q is the elementary charge, Φ_{bm} is the modified effective barrier height between the graphene and the Si, k_B is the Boltzmann constant, T is the temperature, V_a is the applied voltage, and η_{id} is the ideality factor. The tunneling probability α is given by [6]

$$\alpha = \exp\left(-\frac{2t_{ox}\sqrt{2qm^*\Phi_T}}{\hbar}\right), \quad (2)$$

where t_{ox} is the thickness of the insulator layer, m^* is the effective mass of the insulator, Φ_T is the effective barrier height of the insulator layer, and $\hbar = h/2\pi$ is the reduced Planck constant. The modified effective barrier height of the high- κ layer, Φ_{bm} , can be calculated as [7]

$$\Phi_{bm} = \Phi_S - \chi_S + \frac{C_I}{C_I + C_S}\Phi_{GS} + \frac{C_{iI}}{C_{iI} + C_{iS}}\Delta_{IS}, \quad (3)$$

where Φ_S is the work function of the semiconductor; χ_S is the electron affinity of the semiconductor; Φ_{GS} is the difference

between the graphene and semiconductor work functions; Δ_{IS} is the insulator–semiconductor dipole voltage [7]; C_I and C_S are the insulator and semiconductor surface capacitance densities, respectively [7]; and C_{iI} and C_{iS} are the insulator–semiconductor dipole capacitance densities of the insulator and semiconductor, respectively [7]. The theoretically calculated Φ_{bm} of the GIS junctions decreased as the thickness of the high- κ layer was increased (Fig. 2(b)); this occurred because the E_F of Si (5.0 eV) and the charge neutral level (E_{CNL}) of the high- κ materials (Al_2O_3 : 5.33 eV, TiO_2 : 4.6 eV), which have higher energy than the E_{CNL} of Si (4.53 eV), are the main factors affecting the formation of Φ_{bm} when an inserted high- κ layer is present. Increasing the thickness of the high- κ layer shifted E_{FG} down, leading to the formation of a Φ_{bm} lower than that of the control GS junction. Especially, at thicknesses greater than 2-nm high- κ layer, the effective barrier height seems to be almost unchanged because the Fermi level pinning phenomenon to the E_{CNL} of Si is sufficiently alleviated by inserting the thick high- κ layer. In addition, because the inserted high- κ layers have much higher band gaps than the initially formed barrier height, the tunneling probability (α) from graphene to Si shows a decreasing tendency as the thickness of the high- κ layer is increased. Especially, the higher α of the GIS junction with a TiO_2 layer than that of the junction with the Al_2O_3 layer was attributed to the lower valence band offset of TiO_2 (2.03 eV) than that of Al_2O_3 (4.06 eV). Based on these Φ_{bm} and α behaviors, the specific contact resistivity (ρ_c) was reduced dramatically, about 145-fold, when a 1 nm thick high- κ layer was inserted (Fig. 2(c)); this occurred because the effect of the decrease in Φ_{bm} dominated that of the reduced α . Here, the ρ_c was calculated from the J – V characteristic curves using the equation $\rho_c = (\partial J / \partial V)^{-1}|_{V=0}$. However, ρ_c showed an increasing tendency as the high- κ thickness was increased from 1 to 4 nm; it was minimized at 1 nm for Al_2O_3 and 2 nm for TiO_2 . For the high- κ layers thicker than the 2 nm layers, we expect that the reduction in α caused by increasing the thickness further would dominate the effect on ρ_c because Φ_{bm} does not change greatly with thickness in this range.

To verify our theoretical analysis of the GIS junction, we fabricated GIS junction diodes with 1 or 2 nm thick high- κ layers; the insertion of the thin high- κ layers yielded 76-fold, 95-fold, and 273-fold increases in reverse current (at 2 V) for the devices including 1 nm Al_2O_3 , 1 nm TiO_2 , and 2 nm TiO_2 layers, respectively (Fig. 3(a)). Here, since the barristor normally operates in the reverse bias condition, where the reverse current mainly depends on the Φ_{bm} , we focused on the reverse current. In particular, we obtained a significantly improved reverse current density of 2×10^{-1} A/cm² at 2 V by inserting 2-nm TiO_2 layer when compared to those of previously reported GS junction diodes (2×10^{-4} A/cm² [2] and 4×10^{-6} A/cm² [8]) and the graphene barristor (1×10^{-5} A/cm² [5]). The forward currents (under negative voltage) were also increased because ρ_c was reduced by the inclusion of the high- κ layers. The current reduction observed in the 2 nm Al_2O_3 sample seemed to result from the effective reduction in α (due to increased tunneling distance) by increasing the thickness of the Al_2O_3 ,

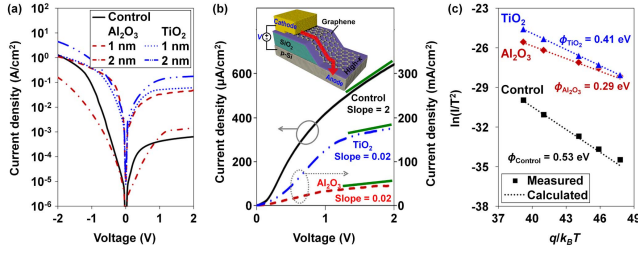


Fig. 3. J - V characteristics of GS and GIS junction diodes in (a) log scale and (b) linear scale. The inset shows the schematic illustration of the GIS junction diode. Here, the red arrow indicates the reverse current direction when a positive V is applied. (c) Richardson plot ($\ln(I/T^2)$) versus $q/k_B T$ for GS and GIS junction diodes at 2 V.

whereas the barrier height lowering was almost stopped. We also note that the forward currents show the slightly different trend from the reverse because η_{id} , which is affected by interfacial state density (D_{it}) in the Si/high- κ layer, as well as Φ_B significantly change the forward current (I_F) that is exponentially dependent on V in the forward bias. However, since the graphene barristor normally operates in the reverse bias condition, the D_{it} does not impact on the performance of the device. The linear scale J - V curves in reverse biased GS and GIS junction diodes are shown in Fig. 3(b). The saturation J - V slope ($=\Delta J/\Delta V$ between 1 V and 2 V) in the GIS junction diodes were reduced (0.02 for 1-nm Al_2O_3 and 0.02 for 2-nm TiO_2), compared to that of GS junction diode (2 for no high- κ), indicating that the dependence of reverse currents on V_{DS} was further weakened by the insertion of high- κ layer. This effect is attributed to the high dielectric constant value of the inserted high- κ layer which finally reduces the degree of barrier height lowering effect. We also performed temperature-dependent electrical measurements of the GIS junction diodes to extract the Φ_{bm} of the barrier between the graphene and the p-Si. Fig. 3(c) shows plots of $\ln(I/T^2)$ versus $1/k_B T$ for 1 nm Al_2O_3 , 2 nm TiO_2 , and control junction diodes; as expected, the respective Φ_{bm} values extracted from these Arrhenius plots were 0.53, 0.29, and 0.41 eV. The theoretical results calculated by using the GIS junction model agreed well with these measured results. Here, because the barrier lowering effect was considered, the lowered Φ_{bm} values were obtained when compared to the calculated Φ_{bm} in Fig. 2(b).

To confirm the degree of ρ_c reduction resulting from the insertion of the high- κ layer, we fabricated MIS junction diodes with Ti, Cr, and Au metals, which have the work functions of 4.3, 4.5, and 5.1 eV, respectively. The normalized ρ_c values (normalized to the ρ_c of the control samples) were reduced in all MS diodes by inserting a 1 nm Al_2O_3 layer; this reduction was greater when 2 nm TiO_2 was used (Fig. 4 (a)). Especially, the improvement in ρ_c yielded by the insertion of a high- κ layer was greater for GIS-junction diodes than for MIS-junction diodes. This is because graphene is highly sensitive to GS interfacial defects due to graphene's finite density of states, which are expected to deteriorate ρ_c ; the highest ρ_c was observed in the control GS junction. Meanwhile, a crucial property of GS or GIS junctions is the ability

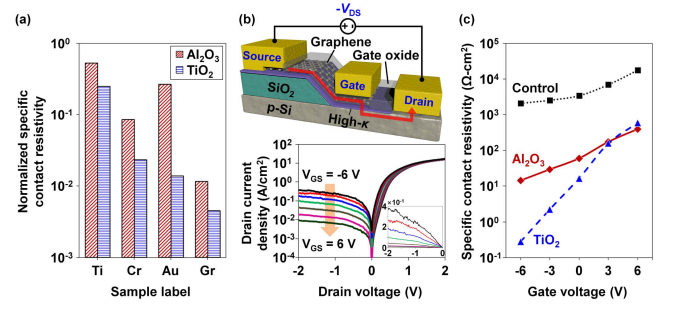


Fig. 4. (a) Normalized specific contact resistivities (ρ_c) of GIS and MIS junctions (1-nm Al_2O_3 or 2-nm TiO_2). The ρ_c values were normalized by dividing the ρ_c of the GIS or MIS junctions by the values of the control junctions (GS or MS). (b) Device structure with V_{GS} -controlled GIS junction and J_D - V_D characteristics of GIS junction-based barristor with a 1-nm thick Al_2O_3 layer under various V_{GS} biases between -6 V and $+6$ V. The inset shows the J_D - V_D curves in linear scale at the reverse bias region. (c) Specific contact resistivity of GS and GIS junctions.

to control the barrier height by modulating E_{FG} by means of V_{GS} . Herein we propose a modified GIGS model that considers gate control of E_{FG} , and also investigate the effect of E_{FG} modulation upon ρ_c , which was calculated based on $\rho_c = (\partial J/\partial V)^{-1}|_{V_{DS}=0}$ at various V_{GS} . Because the shift of E_{FG} resulting from a change in V_{GS} needs to be considered for GIS junctions, unlike MIS junctions, Φ_{GS} is replaced by $\Phi_{GS} - \Delta\Phi_{FG}$ in equation (3), where $\Delta\Phi_{FG}$ is the barrier height variation resulting from a change in V_{GS} at the GIS junction:

$$\Delta\Phi_{FG} = \text{sign}(V_{GS})\hbar v_F \sqrt{\frac{\pi \epsilon_0 \kappa_{Gox} |\Delta V_{GS}|}{q t_{Gox}}}, \quad (4)$$

where v_F is the Fermi velocity of graphene and ϵ_0 is the vacuum permittivity; κ_{Gox} and t_{Gox} are the permittivity and thickness of the gate oxide, respectively. As shown in Fig. 4(b), drain current densities (J_{DS}) of GIS junction-based barristor with a 1-nm thick Al_2O_3 layer are modulated through V_{GS} . Especially, more stable saturation J_{DS} is obtained in the reverse-biased region (negative V_{DS}), where the J_{DS} is less affected by V_{DS} . Fig. 4(c) shows the ρ_c of GS and GIS junctions as a function of V_{GS} calculated by using the modified GIGS model. The Φ_{bm} decreases when a negative V_{GS} is applied because of the down-shift of E_{FG} , consequently reducing ρ_c (or increasing J_{DS}). Oppositely, ρ_c increases (or J_{DS} decreases) under a positive V_{GS} due to the resulting increase in Φ_{bm} . Furthermore, when TiO_2 is inserted, the resulting variation in ρ_c by V_{GS} seems to be greater than that of the control and Al_2O_3 devices. The corresponding barrier height modulation in the TiO_2 sample is about 0.19 eV, which is similar to the previously reported value for the graphene barristor (0.2 eV) [5]. The high κ_{Gox} of TiO_2 (34), which was extracted through capacitance-voltage measurement, induces an increase in C_1 , consequently making the $C_1/(C_1 + C_S)$ term in equation (3) close to 1. Because this corresponds to weak Fermi level pinning at the GIS junction, ρ_c is more effectively changed by V_{GS} .

IV. CONCLUSION

The impact of inserting a high- κ layer between graphene and p-Si was theoretically and experimentally investigated.

Reductions of about 86-fold and 222-fold in the ρ_c of the GIS junctions were achieved when 1 nm Al_2O_3 and 2 nm TiO_2 layers were present, respectively; the corresponding reduced Φ_{bm} values were 0.24 and 0.12 eV. We proposed a GIGS model that simultaneously considered the modulation of E_{FG} by V_{GS} as well as the effect of inserting the high- κ layer (i.e., the E_{FG} shift). This study is expected to contribute to the optimization of various graphene devices based on the concept of the tunable GS junction.

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